

# ASSIGNMENT

1. Find the number of integers in the set  $\{1, 2, 3, \dots, 2009\}$  whose sum of the digits is 11.

**Sol:**

Answer: (133)

We consider 3 sets of integers:

(a) 0001 ~ 0999. We count the number of nonnegative solutions of  $a + b + c = 11$ . It is  ${}^{11+3-1}C_{3-1} = {}^{13}C_2 = 78$ . Note that 11 should not be split as  $0+0+11$  nor  $0+1+10$ . So the number of solutions is  $78 - 3 - 6 = 69$ .

(b) 1001 ~ 1999. We count the number of nonnegative solutions of  $a + b + c = 10$ . It is  ${}^{10+3-1}C_{3-1} = {}^{12}C_2 = 66$ . Note that 10 should not be split as  $0+0+10$ . So the number of solutions is  $66 - 3 = 63$ .

(c) 2001 ~ 2009. We see that only 2009 satisfies the property.

$\therefore$  The total number of integers satisfying the property is  $69 + 63 + 1 = 133$ .

2. A four digit number consists of two distinct pairs of repeated digits (for example 2211, 2626 and 7007). Find the total number of such possible numbers that are divisible by 7 or 101 but no both

**Sol:**

Answer: (97)

We let the 2 distinct digits be  $A$  and  $B$  with  $1 \leq A \leq 9$ ,  $0 \leq B \leq 9$  and  $A \neq B$ .

Case 1:  $ABAB = 101 \times AB$ . There are  $9 \times 9 = 81$  possibilities.

However  $AB$  must not be a multiple of 7 (12 possibilities excluding 07 and 77)  $\Rightarrow$  There are  $81 - 12 = 69$ .

Case 2:  $AABB = 11 \times (100A + B)$ . There are 11 possibilities that are factors of 7.  $(A, B) = (1, 5), (2, 3), (3, 1), (3, 8), (4, 6), (5, 4), (6, 2), (6, 9), (7, 0), (8, 5), (9, 3)$ .

Case 3:  $ABBA = 11 \times (91A + 10B)$ .  $B$  must be 0 or 7. There are 17 possibilities.

$\therefore$  Total such possible number =  $69 + 11 + 17 = 97$ .

3. Using the digits 0,1,2,3 and 4, find the number of 13 digit sequence that can be written so that the difference between any two consecutive digits is 1

Examples of such 13 – digit sequence are 0123432123432, 2323432321234 and 23010101234323

**Sol:**

Answer: (3402)

We call a sequence that satisfy the condition a “good” sequence.

Let  $A_n$  denote the number of “good” sequence that end in either 0 or 4,

$B_n$  denote the number of “good” sequence that end in either 1 or 3,

$C_n$  denote the number of “good” sequence that end in 2.

We have

- (1)  $A_{n+1} = B_n$  because each sequence in  $A_{n+1}$  can be converted to a sequence in  $B_n$  by deleting its last digit.
- (2)  $B_{n+1} = A_n + 2C_n$  because each sequence in  $A_n$  can be converted into a sequence in  $B_{n+1}$  by adding a 1 (if it ends with a 0) or a 3 (if it ends with a 4) to its end, and each sequence in  $C_n$  can be converted into a sequence in  $B_{n+1}$  by adding a 1 or a 3 to it.
- (3)  $C_{n+1} = B_n$  because each sequence in  $C_{n+1}$  can be converted to a sequence in  $B_n$  by deleting the 2 at its end.

Hence we can show that  $B_{n+1} = 3B_{n-1}$  for  $n \geq 2$ .

Check that  $B_1 = 2$  and  $B_2 = 4$ , so  $B_{2n+1} = 2 \times 3^n$  and  $B_{2n} = 4 \times 3^{n-1}$ .

So  $A_{13} + B_{13} + C_{13} = 2B_{12} + B_{13} = 2 \times 4 \times 3^5 + 2 \times 3^6 = 3402$ .

4. There are eight envelopes numbers 1 to 8. Find the number of ways in which 4 identical red buttons and 4 identical blue buttons can be put in the envelopes such that each envelope contains exactly one button, and the sum of the numbers on the envelopes the red buttons is more than the sum of the numbers on the envelopes containing the blue buttons.

**Sol:**

Answer: (D)

There are  $\binom{8}{4} = 70$  ways of putting 4 identical red buttons and 4 identical blue buttons in the envelopes. Since  $1 + 2 + 3 + \dots + 8 = 36$ , there are 8 cases where the sum of the numbers on the envelopes containing the red buttons is equal to 18 (which is also equal to the sum of the numbers on the envelopes containing the blue buttons), namely, (8, 7, 2, 1), (8, 6, 3, 1), (8, 5, 4, 1), (8, 5, 3, 2), (7, 6, 4, 1), (7, 6, 3, 2), (7, 5, 4, 2) and (6, 5, 4, 3). Hence it follows that the required number of ways is  $\frac{70-8}{2} = 31$ .

5. Find the number of positive divisors of  $(2008^3 + (3 \times 2008 \times 2009) + 1)^2$

Sol:

Answer: 91

Let  $a = 2008$ . Then

$$\begin{aligned} (2008^3 + (3 \times 2008 \times 2009) + 1)^2 &= (a^3 + 3a(a+1) + 1)^2 \\ &= (a^3 + 3a^2 + 3a + 1)^2 \\ &= (a+1)^6 = 2009^6 = 7^{12} \cdot 41^6. \end{aligned}$$

Hence the number of positive divisors is  $(12+1)(6+1) = 91$ .

6. Find the number of 2-element subsets  $\{a, b\}$  of  $\{1, 2, 3, \dots, 99, 100\}$  such that  $ab + a + b$  is a multiple of 7.

Sol:

Answer: 602

Note that  $ab + a + b = (a + 1)(b + 1) - 1$ . Thus  $ab + a + b$  is a multiple of 7 if and only if  $(a + 1)(b + 1) \equiv 1 \pmod{7}$ .

Let  $A = \{1, 2, 3, \dots, 99, 100\}$ , and let  $A_i = \{x \in A : x \equiv i \pmod{7}\}$  for  $i = 0, 1, 2, \dots, 6$ . It is easy to verify that for any  $x \in A_i$  and  $y \in A_j$ , where  $0 \leq i \leq j \leq 6$ ,  $xy \equiv 1 \pmod{7}$  if and only if  $i = j \in \{1, 6\}$ , or  $i = 2$  and  $j = 4$ , or  $i = 3$  and  $j = 5$ .

Thus we consider three cases.

Case 1:  $a + 1, b + 1 \in A_i$  for  $i \in \{1, 6\}$ .

Then  $a, b \in A_i$  for  $i \in \{0, 5\}$ . As  $|A_0| = |A_5| = 14$ , the number of such subsets  $\{a, b\}$  is  $2 \binom{14}{2} = 182$ .

Case 2:  $a + 1$  and  $b + 1$  are contained in  $A_2$  and  $A_4$  respectively, but not in the same set.

Then  $a$  and  $b$  are contained in  $A_1$  and  $A_3$  respectively, but not in the same set.

Since  $|A_1| = 15$  and  $|A_3| = 14$ , the number of such subsets  $\{a, b\}$  is  $14 \times 15 = 210$ .

Case 3:  $a + 1$  and  $b + 1$  are contained in  $A_3$  and  $A_5$  respectively, but not in the same set.

Then  $a$  and  $b$  are contained in  $A_2$  and  $A_4$  respectively, but not in the same set.

Note that  $|A_2| = 15$  and  $|A_4| = 14$ . Thus the number of such subsets  $\{a, b\}$  is  $14 \times 15 = 210$ .

Hence the answer is  $182 + 2 \times 210 = 602$ .

7. Find the number of 0–1 binary sequences for by six 0's and six 1's such that no three 0's are together. For example, 110010100101 is such a sequence but 101011000101 and 11010110001 are not

**Sol**

Answer: 357

First put the six 1's in one sequence. Then there are 7 gaps before the first 1, between two adjacent 1's and after the last 1. For each such gap, we can put a single 0 or double 0's (that is, 00).

If there are exactly  $i$  double 0's, then there are exactly  $6 - 2i$  single 0's, where  $i = 0, 1, 2, 3$ . Therefore the number of such binary sequences with exactly  $i$  double 0's is  $\binom{7}{i} \binom{7-i}{6-2i}$ . Hence the answers is  $\sum_{i=0}^3 \binom{7}{i} \binom{7-i}{6-2i} = 357$ .

8. In each of the following 6-digit positive integers 555555, 555333, 818811, 300388 every digit in the number appears at least twice. Find the number of such 6-digit positive integers.

**Sol.**

Answer: 11754

First note that if every digit in the 6-digit number appears at least twice, then there cannot be four distinct digits in the number. In other words, the number can only be formed by using one digit, two distinct digits or three distinct digits respectively. Therefore we consider three cases.

Case 1: The 6-digit number is formed by only one digit.

Then the number of such 6-digit numbers is clearly 9.

Case 2: The 6-digit number is formed by two distinct digits.

First, the number of such 6-digit numbers formed by two given digits  $i$  and  $j$ , where  $1 \leq i < j \leq 9$ , is

$$\binom{6}{2} + \binom{6}{3} + \binom{6}{4} = 50.$$

Next, the number of such 6-digit numbers formed by 0 and a given digit  $i$ , where  $1 \leq i \leq 9$ , is

$$\binom{5}{2} + \binom{5}{3} + \binom{5}{4} = 25.$$

Therefore the total number of such 6-digit numbers formed by two distinct digits is

$$\binom{9}{2} \times 50 + 9 \times 25 = 2025.$$

Case 3: The 6-digit number is formed by three distinct digits.

First, the number of such 6-digit numbers formed by three given digits  $i, j$  and  $k$ , where  $1 \leq i < j < k \leq 9$ , is

$$\binom{6}{2} \cdot \binom{4}{2} = 90.$$

Next, the number of such 6-digit numbers formed by 0 and two given digits  $i$  and  $j$ , where  $1 \leq i < j \leq 9$ , is

$$\binom{5}{2} \cdot \binom{4}{2} = 60.$$

Therefore the total number of such 6-digit numbers formed by three distinct digits is

$$\binom{9}{3} \times 90 + \binom{9}{2} \times 60 = 9720.$$

Hence the answer is  $9 + 2025 + 9720 = 11754$ .

9. Using the digits 1,2,3,4,5,6,7,8, we can form  $8! (=40320)$  8-digit numbers in which the eight digits are all distinct. For  $1 \leq k \leq 40320$ , let “a” denote the  $k$ th number if these numbers are arranged in increasing order:

12345678, 12345687, 12345768, .....87654321 ;

that is  $a_1 = 12345678$ ,  $a_2 = 1245687$ , .... $a_{40320} = 87654321$ . Find  $a_{2009} - a_{2008}$

**Sol**



Answer: 441

First we determine  $a_{2008}$  and  $a_{2009}$ . Suppose that  $a_{2008} = \overline{x_1x_2x_3x_4x_5x_6x_7x_8}$ , where the  $x_i$ 's are distinct digits in  $\{1, 2, 3, 4, 5, 6, 7, 8\}$ .

Let  $A = \{a_k : k = 1, 2, \dots, 40320\}$ .

Since  $7! = 5040 > 2008$ , we deduce that  $x_1 = 1$ , as there are more than 2008 numbers in  $A$  such that the first digit is 1.

As  $2 \times 6! < 2008 < 3 \times 6!$ , we have  $x_2 = 4$ , as there are less than 2008 numbers in  $A$  such that the first digit is 1 and the second digit is 2 or 3, but there are more than 2008 numbers in  $A$  such that the first digit is 1 and the second digit is 2, 3 or 4. Similarly, since  $2 \times 6! + 4 \times 5! < 2008 < 2 \times 6! + 5 \times 5!$ , we see that the third digit  $x_3$  is 7. By repeating the argument and using the inequalities

$2 \times 6! + 4 \times 5! + 3 \times 4! < 2008 < 2 \times 6! + 4 \times 5! + 4 \times 4!$  and

$2004 = 2 \times 6! + 4 \times 5! + 3 \times 4! + 2 \times 3! < 2008 < 2 \times 6! + 4 \times 5! + 3 \times 4! + 3 \times 3!$ , we obtain  $x_4 = 6$ ,  $x_5 = 5$ . Note also that among the numbers in  $A$  of the form

1476\*\*\*\*, the digit 5 first appears as the fifth digit in  $a_{2008}$  if the numbers are arranged in increasing order. Consequently, as the last three digits are 2, 3 and 8, we must have  $a_{2008} = 14765238$ . It follows that  $a_{2006} = 14765283$ ,

$a_{2007} = 14765328$ ,  $a_{2008} = 14765382$ , and  $a_{2009} = 14765823$ . Hence

$a_{2009} - a_{2008} = 14765823 - 14765382 = 441$ .

10. Let  $A$  be an  $n$ -element subset of  $\{1, 2, \dots, 2009\}$  with the property that the difference between any two numbers in  $A$  is not a prime number. Find the largest possible value of  $n$ . Find a set with this number of elements. (Note : 1 is not a prime number)

Sol.

3. If  $n \in A$ , then  $n + i \notin A$ ,  $i = 2, 3, 5, 7$ . Among  $n + 1, n + 4, n + 6$  at most one can be in  $A$ . Thus among any 8 consecutive integers, at most 2 can be in  $S$ . Hence  $|A| \leq 2 \lceil 2009/8 \rceil = 504$ . Such a set is  $\{4k + 1 : k = 0, 1, \dots, 502\}$ .

$\{1, 5, 9, \dots, 2009\}$

11. Let  $S = \{1, 2, 3, \dots, 30\}$ . Determine the number of vectors  $(x, y, z, w)$  with  $x, y, z, w \in S$  such that  $x < w$  and  $y < z < w$ .

Sol

Answer: 90335

There are two cases to consider: Case (1)  $x \in \{y, z\}$  and Case (2):  $x \notin \{y, z\}$ . For Case (1), there are  $2 \binom{30}{3}$  ways and for Case (2), there are  $3 \binom{30}{4}$  ways. Hence, total number of ways = 90335.

12. Let  $f(n)$  be the number of 0's in the decimal representation of the positive integer  $n$ . For example  $f(10001123) = 3$  and  $f(1234567) = 0$ . Find the value of  $f(1) + f(2) + f(3) + \dots + f(99999)$

**Sol.**

Answer: 38889

Let  $S = \{1, 2, 3, \dots, 99999\}$ , and  $S_i = \{n \in S: f(n) = i\}$  for  $i \geq 0$ . Thus,

$$S = \bigcup_{0 \leq i \leq 4} S_i.$$

For  $0 \leq i \leq 4$ , if  $n \in S_i$ , and  $n$  has exactly  $k$  digits in the decimal representation, then exactly we have  $k - i$  digits are non-zero. Thus,

$$|S_i| = \sum_{k=i+1}^5 \binom{k-1}{i} 9^{k-i}.$$

Then, it is clear that

$$M = |S_1| + 2|S_2| + 3|S_3| + 4|S_4| = 38889.$$

**13.** Let  $S = \{1, 2, 3, 4, \dots, 16\}$ . In each of the following subsets of  $S$ .

$\{6\}$ ,  $\{1, 2, 3\}$ ,  $\{5, 7, 9, 10, 11, 12\}$ ,  $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$

The sum of the all the elements is a multiple of 3. Find the total number of non empty subsets  $A$  of  $S$  such that the sum of all elements in  $A$  is a multiple of 3.

**Sol.**

Answer: 21855.

Let  $S_i = \{x \in S: x \equiv i \pmod{3}\}$  for  $i = 0, 1, 2$ . Note that  $|S_0| = 5$ ,  $|S_1| = 6$  and  $|S_2| = 5$ . Let  $\partial$  be the set of all subsets  $A$  of  $S$  such that  $\sum_{x \in \partial} x$  is a multiple of 3. Note that for any  $A \subseteq S$ ,

$$\sum_{x \in A} x = \sum_{i=0}^2 \sum_{x \in A \cap S_i} x \equiv |A \cap S_1| + 2|A \cap S_2| \pmod{3}.$$

Thus,  $A \in \partial$  if and only if  $|A \cap S_1| \equiv |A \cap S_2| \pmod{3}$ . Thus, it is clear that

**14.** Determine the number of ways of tilling a  $4 \times 9$  rectangle by tiles of size  $1 \times 2$ .

**Sol.**

Let  $f_n$  be the number of ways of tiling a  $4 \times n$  rectangle. Also, let  $g_n$  be the number of ways of tiling a  $4 \times n$  rectangle with the top or bottom two squares in the last column missing, and let  $h_n$  be the number of ways of tiling a  $4 \times n$  rectangle with the top and bottom squares in the last column missing. Set up a system of recurrence relations involving  $f_n$ ,  $g_n$  and  $h_n$  by considering the ways to cover the  $n$ th column of a  $4 \times n$  rectangle. If two vertical tiles are used, then there are  $f_{n-1}$  ways. If one vertical tile and two adjacent horizontal tiles are used, then there are  $2g_{n-1}$  ways. If one vertical tile and two non-adjacent horizontal tiles are used, then there are  $h_{n-1}$  ways. If four horizontal tiles are used, then there are  $f_{n-2}$  ways. Similarly, one can establish the recurrence relations for  $g_n$  and  $h_n$ . In conclusion, we obtain for  $n \geq 2$ ,

$$\begin{aligned} f_n &= f_{n-1} + f_{n-2} + 2g_{n-1} + h_{n-1} \\ g_n &= g_{n-1} + f_{n-1} \\ h_n &= h_{n-2} + f_{n-1}, \end{aligned}$$

With initial conditions  $f_0 = f_1 = g_1 = h_1 = 1$  and  $h_0 = 0$ . Solving  $f_n$  recursively, we obtain  $f_9 = 6336$ .

15. Find the number of 7 digit positive integer such that the digits from left to right non increasing (Examples of 7 digit non increasing numbers are 9998766 and 555555 ; An example of a number that is NOT non increasing is 7776556).

**Sol**

Answer: 11439

Each 16-digit binary sequence containing exactly nine '0's and seven '1's can be matched uniquely to such a 7-digit integer or 0000000 as follows: Each '1' will be replaced by a digit from 0 to 9 in this way: the number of '0's to the right of a particular '1' indicates the value of the digit. For example, 0110000010101101  $\sim$  8832110 and 1111000000000111  $\sim$  9999000.

Thus, required number =  $\binom{16}{7} - 1$ .